

HOUSTON CITY COLLEGE

NEURAL NETWORK ZOO

A Study of Learning Architectures Through Living Systems



TEAM CONTRIBUTION RECORD

THE CIRCLE OF LIFE | THE MATHEMATICAL CORE

KNOWLEDGE THROUGH STRUCTURE

Each architectural model represents a different strategy for transforming information. Together, the examples form a single ecosystem of learning rather than isolated algorithms.

TEAM MEMBER	CLAIMED SECTION	WHAT I DID	WHY I CHOSE IT
Olu	The Learning Cycle & Acacia/Flora (FNN/MLP)	I summarized the first part of the video. I wrote down the four steps of the learning cycle (Input, Process, Output, Feedback) and explained that the Flora model is basically just a forward-only process where information moves layer by layer.	I grabbed the first few slides to establish the foundational definitions early on. It seemed like the most straightforward starting point before the architectures got too complicated.
Cherluk	Chameleon (CNN) & Timber Wolf (LSTM/RNN)	I handled the middle slides. I documented that the Chameleon model learns visual patterns from pixels to shapes, and the Timber Wolf model keeps past information in its memory to influence the next prediction across time.	I found the spatial feature extraction and sequence memory concepts to be the most interesting parts of the video, and taking these two animals neatly covered the middle chunk of the work.
Taki	Mycelium/Fungus (Autoencoder) & Summary	I covered the compression model, explaining how Autoencoders compress and reconstruct data. I also typed up the notes on why the four examples matter to different data types.	I volunteered for the last animal analogy because I wanted to handle the concluding thoughts. It made the most sense for me to just tie all the visual analogies together at the end.
Michael	“Circle of Life” & The Mathematical Core	I explained that all four models can operate in the same world even if their internal strategies change. I also documented the underlying math behind the visual metaphors, specifically adding the artificial neuron equation $y = f\left(\sum wx + b\right)$ to our notes.	I prefer focusing on the actual numerical mechanisms and technical architecture behind the machine learning models rather than just sticking to the visual teaching devices, so I wanted to make sure our notes included the actual math.

MATHEMATICAL CORE

The visual metaphor changes; the numerical foundation remains.

$$y = f\left(\sum wx + b\right)$$